RC-QASyn: Auditable Evidence, Coverage-Aware Construction and Bounded Repair for Synthetic Tabular Data

Dev Sangam Kumar
Department of Artificial Intelligence
Sardar Vallabhbhai National Institute of Technology
Surat, India
u24ai070@aid.svnit.ac.in

Aayush Pachpute
Department of Artificial Intelligence
Sardar Vallabhbhai National Institute of Technology
Surat, India
u24ai097@aid.svnit.ac.in

Manan Jariwala
Department of Artificial Intelligence
Sardar Vallabhbhai National Institute of Technology
Surat, India
u24ai073@aid.svnit.ac.in

Priyanka Bhatt
Reviewer and Supervisor
Department of Artificial Intelligence
Sardar Vallabhbhai National Institute of Technology
Surat, India
priyankabhatt2133@gmail.com

***Abstract*—Among 601 historical LLM-generated tabular candidates from three imbalanced classification datasets, 75 (12.5%) violate source-declared structural requirements. RC-QASyn keeps structural compliance, statistical support and dataset contribution separately auditable through provenance, schema checks, local and global TRAIN evidence, descriptive admission, fixed-budget construction and bounded integer repair. All 54 eligible repairs become structurally compliant. At matched budgets, quality-coverage selection raises distinct-feature TRAIN coverage over source order from 0.622 to 0.948 on Abalone19 and from 0.416 to 0.506 on Ecoli4, with zero exact duplicates. Better geometry does not ensure better classification: expanding the repaired page-blocks0 pool lowers logistic pure-synthetic F1 from 0.446 to 0.342 (paired conditional 95% difference interval $[-0.135, -0.074]$). On five fresh train-only splits per dataset, curated augmentation brings no uniform benefit over real-only training, and random curation remains competitive. An exploratory addendum with six more KEEL datasets, re-implemented learned curators and four published oversamplers reproduces the coverage gain without a consistent F1 or average-precision advantage. These results establish reproducible structural and geometric effects with mixed utility outcomes, without independently establishing complete semantic validity.**

***Index Terms*—synthetic tabular data; curation; coverage; bounded repair; imbalanced learning**

I. Introduction

A generated record can be readable and numerically plausible while omitting its class, violating an integer declaration or duplicating another record. In the historical population studied here, 75 of 601 candidates violate authoritative structure, including 64 of 200 page-blocks0 records. These rates describe one generation campaign, not LLM output in general, but they motivate checking record content before interpreting resemblance or usefulness.

Three questions stay distinct: does a candidate comply with justified source declarations, how well does it resemble TRAIN locally and globally, and what does it contribute to a selected dataset and its classifier? A distant record may be legitimate, a compliant one redundant, and a diverse subset may train a worse classifier. We ask how far structural filtering, heterogeneous evidence, coverage-aware selection and bounded repair improve measurable data characteristics and classification utility compared with simple alternatives, using a controlled historical audit and a separate offline replication. The contributions are:

1) auditable separation of schema compliance, local/global statistical support and dataset contribution, linked to immutable candidate identities;
2) evidence-aware fixed-budget construction compared with random, source-order, farthest-first and coverage-greedy controls;
3) source-preserving bounded integer repair with controlled reevaluation;
4) an analysis of where geometry and classification agree and disagree, with paired prediction uncertainty and split-specific offline generation.

These are integration and empirical contributions built from established components; no new optimality guarantee is claimed. Annotation experiments and validity calibrators are outside the scope.

II. Related work

Generation and curation. CTGAN, TabDDPM and GReaT generate tabular data with adversarial, diffusion and language models [1], [2], [3], and EPIC prompts LLMs for imbalanced-class synthesis [4]. RC-QASyn instead curates an existing candidate stream. Closer work filters records after generation: CLLM keeps high-confidence, low-aleatoric rows under a classifier's learning dynamics [5], ReFine suppresses local

redundancy while keeping rare records [6], Team-then-Trim adds a three-stage quality-control pipeline [7], and RUBRIC ranks oversampled minority candidates by learned density-ratio realism and boundary utility [8]. RC-QASyn differs in using model-free support evidence (TRAIN-scaled nearest-neighbour and shrinkage Mahalanobis distances), explicit reject, abstain and review outcomes, and source-declared structural checks with bounded repair. We do not claim it outperforms learned curation; the addendum compares re-implemented principles of CLLM and RUBRIC, not the original systems (Section V-F).

Evaluation, coverage and selection. Sample-level evaluation separates fidelity, diversity and generalization [9], and density and coverage metrics define support by k-nearest-neighbour balls [10]; our fixed-radius TRAIN coverage is closely related, and our duplication rates carry no probabilistic or privacy meaning. Greedy coverage and diversity selection are well studied [11], [12], and a well-chosen synthetic subset can outperform the full set [13]; classical greedy guarantees do not automatically extend to our lexicographic selector. Because quality filtering can remove modes [14], coverage stays an explicit objective. Candidate percentiles are not calibrated in the sense of [15]; bootstrap inference depends on its resampling design [16], Wilson intervals address small positive counts [17], and surveys and LLM-judge studies limit what recorded critic decisions establish [18], [19].

Validation, repair and oversampling. Schema validation is established practice in ML pipelines [20]. Constraint-aware models enforce constraints during or after generation [21], and recent work learns inter-column constraints and repairs generator output [22]. RC-QASyn checks only source-justified declarations and allows a narrow integer-only correction with an explicit edit budget, so it never fabricates labels or features. SMOTE interpolates minority observations [23]; Tomek links, edited nearest neighbours [24] and SMOTE-IPF [25] clean after oversampling, close in spirit to support-based admission. We use imbalanced-learn, including SMOTENC [26]. Oversampling, random selection and class weighting answer different questions: generation, curation control and classifier training.

III. Methodology

Let T denote REAL TRAIN, E REAL TEST and $G = \{(o_i, x_i, y_i)\}$ the candidate stream. Occurrence identity distinguishes physical records with equal payloads, and source files, ordinals, payload hashes, specifications and experiment identities link each selected record to its inputs. A missing class stays missing; generation conditioning never supplies it.

A. Structure and statistical evidence

Specifications declare required features and class, numerical primitives, nominal values, integer fields and permitted fields. Supported violations can reject; observed ranges are not treated as physical bounds. For p features, the TRAIN-scaled mixed distance is

$$d(x,t) = \frac{1}{p}\sum_j \delta_j(x_j, t_j), \qquad \delta_j = \begin{cases} \min(|x_j - t_j|/R_j, 1), & \text{numeric, } R_j > 0, \\ \mathbf{1}[x_j \neq t_j], & \text{nominal or } R_j = 0, \end{cases} \tag{1}$$

where R_j is the TRAIN span and class is excluded. Local support $L_k(x)$ averages distances to the nearest $\min(k, |T_s|)$ supported TRAIN observations (ties broken by row identity; primary $k = 5$). Global consistency scales numerical features by TRAIN median/IQR (falling back to span, then one), one-hot encodes TRAIN categories and, for encoded dimension D, covariance S and mean μ, uses

$$\Sigma = (1-\alpha)S + \alpha\,\mathrm{diag}(S) + \lambda I, \qquad Q(x) = \sqrt{\max\big(0, (z-\mu)^\top \Sigma^{-1}(z-\mu)\big)/D}, \tag{2}$$

with primary $\alpha = .10$, $\lambda = 10^{-8}$. Unsupported evidence and unseen categories stay explicit.

B. Descriptive combination and admission

For the reference set A_f of candidate scores in a dataset and evidence family,

$$u_f(s) = \big(\#\{a < s\} + \tfrac{1}{2}\#\{a = s\}\big)/|A_f|, \tag{3}$$

where larger ranks mean weaker support. Combined concern $\max(u_L, u_Q)$ requires both components and is an uncalibrated index, not a validity probability; no independence assumption or learned weight is used. Admission rejects authoritative defects, abstains when required evidence is missing, requests review for partial structure or concern of at least $q = .90$, and otherwise provisionally passes. Structure, local and global policies form ablations, and $q = .80, .95$ are historical sensitivities. Track A keeps historical models and ranks, including for derived queries; Track B recomputes them for each new TRAIN-fitted pool.

C. Coverage-aware construction

Let $\mathcal{R}$ be the distinct TRAIN feature/class regions and $h_t(S) = \mathbf{1}[\min_{s\in S} d(s,t) \leq .05]$. Uniform coverage averages h_t over regions, balanced coverage weights each class-y region by $1/(Kn_y)$, and distinct-feature TRAIN coverage is reported separately. At budget b, source order takes the earliest occurrences, farthest-first maximizes minimum separation and coverage-greedy maximizes uniform marginal coverage. Quality-coverage forbids repeated feature payloads, first fills a feasible minority quota $\lfloor .5b \rfloor$, and then ranks lexicographically by balanced marginal coverage, fewer neighbours within .01, greater separation, lower concern and source order. Exact redundancy is $1 - u/n$ for u unique payloads among n selections; near redundancy is the fraction of selections with another within .01.

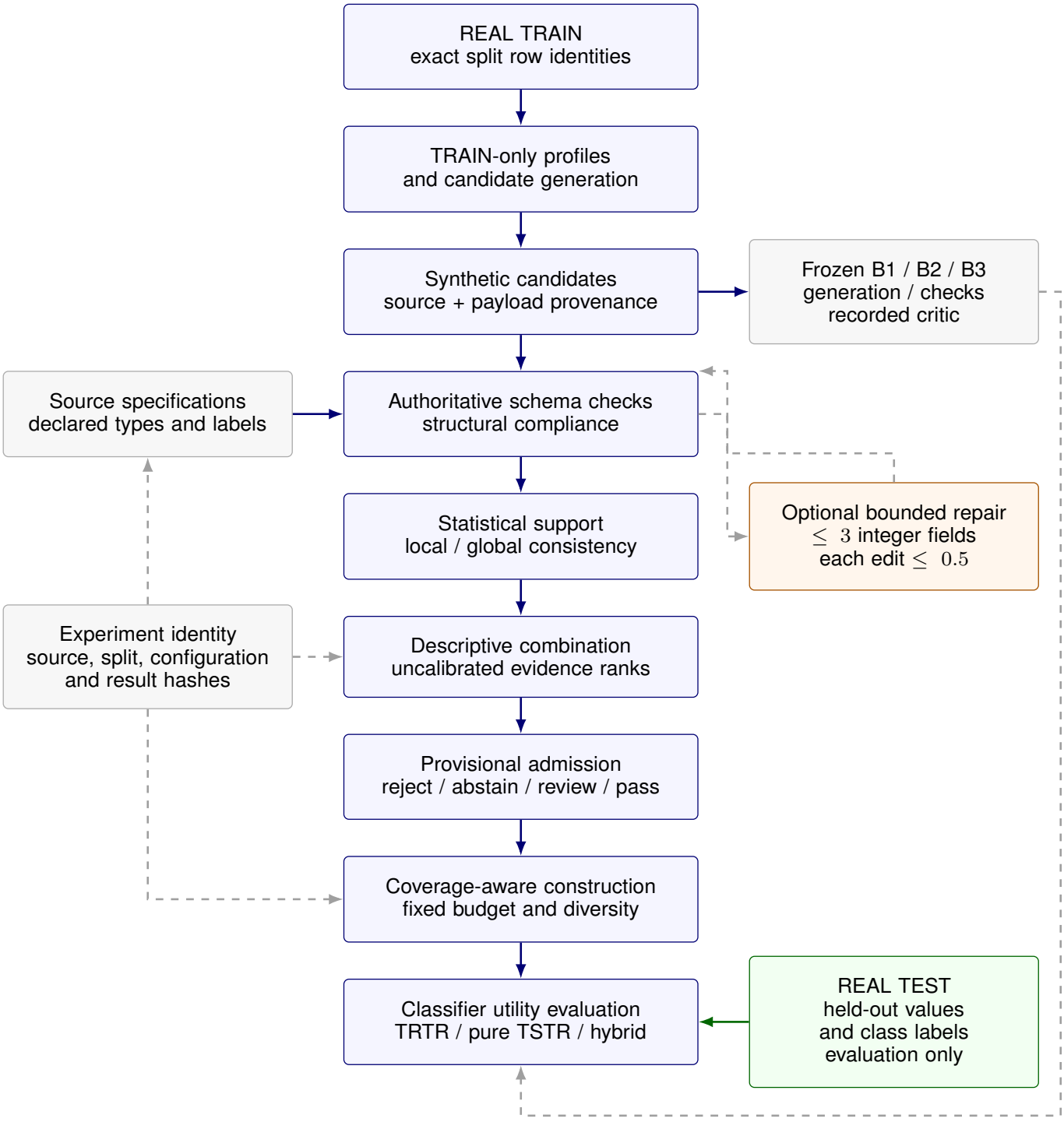


Fig. 1. RC-QASyn architecture. Structure and statistical support are separate; optional repair creates derived versions and repeats checks. B1/B2/B3 remain frozen controls, and REAL TEST enters only evaluation. Dashed arrows denote provenance or optional reevaluation.

D. Bounded structural repair

Only fractional finite values in source-declared integer fields can change. Decimal half-to-even rounding must satisfy

$$|J| \le 3, \quad |x'_j - x_j| \le .5 \ (j \in J), \qquad x'_j = x_j \ (j \notin J). \tag{4}$$

Missing targets or features, category substitution and imputation are excluded, and a record needing more than three edits is refused. Derived identities bind parent, policy, specification and payload; originals stay immutable, frozen evidence and admission rescore repairs, and parent exclusivity prevents double counting. Expanded-pool comparisons therefore measure

TABLE I
FROZEN TASK DATASETS AND LITERAL POSITIVE COUNTS.

Dataset	Rows	TRAIN	TEST	TRAIN+	TEST+	Candidates
Abalone19	4174	3340	834	26	6	200
Ecoli4	336	269	67	16	4	201
page-blocks0	5472	4377	1095	447	112	200

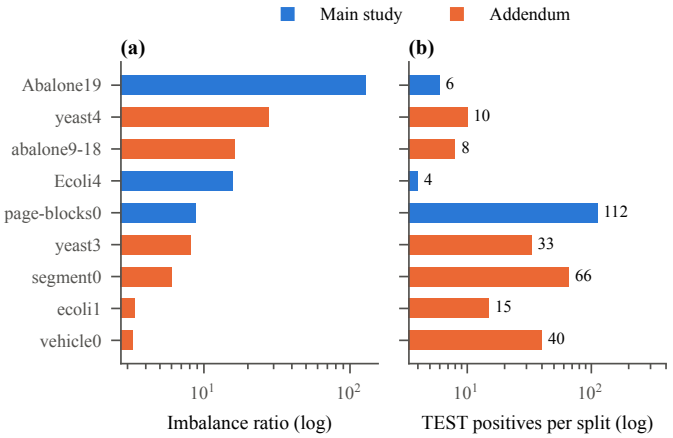


Fig. 2. Datasets of the main study and the exploratory addendum (Section V-F). (a) Imbalance ratio. (b) TEST positives per fresh split, which bound inference on the smallest datasets.

repair plus reselection, not the isolated effect of one edit.

IV. EXPERIMENTAL DESIGN

The protocol was committed before any new score was computed (`91619f1`) on frozen main `244a9bf`, and settings stay fixed after scoring. The research questions cover structural frequency (RQ1), evidence complementarity (RQ2), coverage and redundancy (RQ3), repair (RQ4), utility (RQ5) and stability and uncertainty (RQ6).

Track A (historical fixed split). Datasets, original TRAIN/TEST identities and the 601 candidates are unchanged; candidates depended on original TRAIN and are never reassigned, the previously evaluated TEST set is reused descriptively, and exact TRAIN/TEST feature overlap is recorded. Budgets of 56, 70 and 49 for Abalone19, Ecoli4 and page-blocks0 are inherited. Raw-labeled random keeps evaluable integer defects but excludes absent labels and unsupported features, schema random requires full compliance, and admitted random draws from the combined-admitted pool shared by the four deterministic selectors; random controls use seeds $13, 29, 43, 61, 79$. Admission ablations apply the same quality-coverage selector after structure-only, local, global or combined eligibility, so pool size is part of the intervention. All 138 historical utility comparisons are replayed from saved predictions; B1/B2/B3 stay frozen and are intersected with structural completeness. The archived raw scorer could read a missing class as the string `nan`, so those scores are not used as complete binary-record comparisons. External comparators append exactly b minority rows with ROS or SMOTE (SMOTENC for categorical Abalone19), with TRAIN-fitted standardization and unrounded page-blocks0 integers, and balanced logistic loss is a non-generative comparator. These are training alternatives, not same-pool curation; mixed-class historical subsets may add fewer than b minority rows in hybrid regimes.

Track B (fresh offline splits). Five deterministic stratified 80/20 splits per dataset use the same seeds, and every split refits preprocessing, generation, evidence and geometry on its own TRAIN. Pools hold $4b$ minority records from SMOTENC (Abalone19), SMOTE (Ecoli4) or ROS (integer-declared page-blocks0, which adds no new support), with $k = \min(5, n_+ - 1)$. Six curators (schema random, admitted random, source order, farthest-first, coverage-greedy, quality-coverage) are compared with real-only, class-weighted and ROS/SMOTE training. Minority-only pools allow hybrid augmentation but not pure TSTR. This tests offline curation, not LLM candidate quality, and because holdouts overlap, across-split SD describes variability, not an independent-sample standard error.

Classifiers, metrics and uncertainty. Logistic regression (liblinear, at most 300 iterations) and random forest (50 trees) use seed 42, with imputation, scaling and encoding fitted only on the training population. TRTR trains on real TRAIN, pure TSTR on synthetic rows and hybrid on real TRAIN plus synthetic positives. We report $F_1 = 2TP/(2TP + FP + FN)$, balanced accuracy, MCC, AP (the step integral $\sum_k (R_k - R_{k-1})P_k$, not trapezoidal area) and AUROC; point MCC uses the frozen zero convention for degenerate margins, and undefined cases are counted explicitly. Historical intervals use 2,000 stratified paired resamples of saved predictions (seed 20261025, 95% percentile bounds), resampling positive and negative indices within strata identically across paired methods. Primary pairs compare original quality-coverage with schema and admitted random (seed 13), hybrid quality-coverage with TRTR, and repaired with original quality-coverage. Intervals condition on fitted models and observed TEST, excluding generation and refitting uncertainty. No multiplicity-adjusted significance claim is made.

V. RESULTS

A. RQ1–RQ2: structure and complementary support

There are 75 structural rejects (12.48%): five of 200 Abalone19, six of 201 Ecoli4 and 64 of 200 page-blocks0 candidates (Fig. 3a). Fifteen lack class, one lacks features and 64 have fractional-integer fields; 26 violate multiple fields, and no unsupported nominal value or invalid numerical primitive appears. Local and global evidence are related but flag different records (Fig. 3b–d): Spearman correlations are .908, .488 and .621 for Abalone19, Ecoli4 and page-blocks0, with 14, 30 and two tail disagreements. Combined admission yields 170, 162 and 119 provisional passes, against 195, 195 and 136 with structure only. This shows distinct statistical information, not semantic-error detection performance.

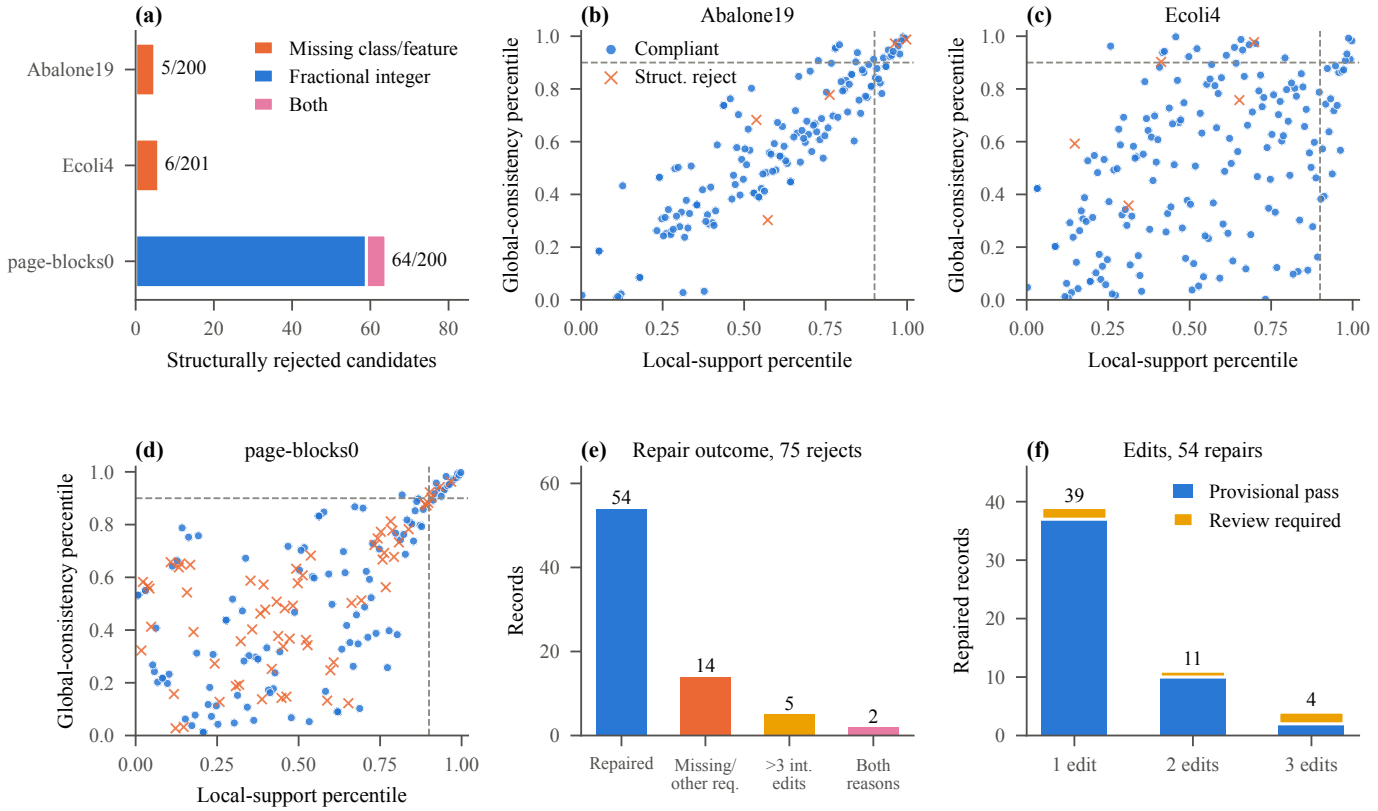


Fig. 3. Historical candidate audit. (a) Structural rejects by violation type. (b–d) Local-support and global-consistency percentiles of every candidate with available evidence; dashed lines mark the review cutoff $q = .90$, crosses mark structural rejects. (e) Repair outcome for the 75 rejects. (f) Integer edits per repaired record and admission after rescoring.

B. RQ3: matched-budget ablations

Every construction reaches its budget (Table II, Fig. 4). Quality-coverage raises distinct-feature TRAIN coverage over source order from .6225 to .9479 on Abalone19 and from .4164 to .5056 on Ecoli4, with zero exact and near redundancy. Farthest-first also removes redundancy but covers less of Ecoli4 (.4684); coverage-greedy matches .5056 with exact duplication .1143. Eligibility, quotas and coverage act together, so this is not the isolated effect of an evidence score. Additional evidence also changes outcomes non-monotonically: structure-only admission slightly exceeds full-admission coverage on Abalone19 (.951 versus .948) and Ecoli4 (.509 versus .506), whereas on page-blocks0 full admission raises logistic pure-TSTR F1 from .202 to .446.

C. RQ4: bounded repair

All 54 eligible repairs become compliant; 21 rejects are ineligible (Fig. 3e), including five page-blocks0 records that also lack class and five that need more than three edits. Repairs edit 73 fields: 39 records need one edit, eleven two and four three, with absolute changes of .1–.5 (mean .3119). Under unchanged admission, 49 repaired versions provisionally pass and five need review, so the page-blocks0 admitted pool grows from 119 to 168. At budget 49, quality-coverage selects 16 repaired records: coverage rises from .9324 to .9370 and near duplication falls from .3265 to .1020. Source-order coverage declines slightly, so expanded pools have method-dependent effects.

D. RQ5: utility and external comparators

All 138 historical utility rows reproduce within 10^{-12}. The expanded study has 894 utility rows, 678 successful and 216 explicitly unavailable, with no execution failures or budget shortfalls. The external baselines (Table III) are training alternatives, not matched curation. Standard SMOTE output for page-blocks0 is not integer-coerced, so 46 of 49 generated rows fail the integer contract while remaining evaluable numerical augmentation.

Geometry and utility can disagree. For page-blocks0 logistic pure TSTR, repair-pool expansion lowers F1 from .4464 to .3421, while AP moves from .5552 to .5716 and hybrid F1 stays .7150, despite the coverage and redundancy gains of RQ4.

E. RQ6: conditional uncertainty and fresh-split stability

Saved predictions give 216 endpoint and 168 paired-difference intervals (Fig. 5). The page-blocks0 logistic repair-minus-original F1 difference is about $-.1043$ (interval $[-.1350, -.0739]$) and the AP difference .0165 ($[-.0071, .0493]$): F1 decreases consistently under this design, while the AP direction is less stable. Small positive counts

TABLE II

CONSOLIDATED HISTORICAL ABLATIONS. Q DENOTES UNCHANGED QUALITY-COVERAGE SELECTION. RANDOM ROWS AVERAGE FIVE SEEDS; COMPLETE PER-SEED DATA ACCOMPANY THE PAPER. F1 IS LOGISTIC PURE TSTR; – MEANS UNAVAILABLE.

Dataset	Method	Pool	n	Coverage	Exact	Near	F1
Abalone19	Raw labeled random	195	56	0.799	0.171	0.443	0.029
	Schema random	195	56	0.799	0.171	0.443	0.029
	Admitted random	170	56	0.833	0.236	0.493	0.022
	Structure + Q selection	195	56	0.951	0.000	0.000	0.030
	Structure/local + Q	177	56	0.950	0.000	0.000	0.031
	Structure/global + Q	177	56	0.948	0.000	0.000	0.031
	Structure/local/global + Q	170	56	0.948	0.000	0.000	0.031
	Admitted source order	170	56	0.622	0.357	0.625	–
	Admitted farthest first	170	56	0.940	0.000	0.000	0.035
	Admitted coverage greedy	170	56	0.948	0.107	0.268	0.030
	Repair pool + Q	170	56	0.948	0.000	0.000	0.031
Ecoli4	Raw labeled random	195	70	0.422	0.097	0.180	0.253
	Schema random	195	70	0.422	0.097	0.180	0.253
	Admitted random	162	70	0.439	0.117	0.197	0.217
	Structure + Q selection	195	70	0.509	0.000	0.000	0.216
	Structure/local + Q	175	70	0.509	0.000	0.000	0.195
	Structure/global + Q	177	70	0.506	0.000	0.000	0.229
	Structure/local/global + Q	162	70	0.506	0.000	0.000	0.205
	Admitted source order	162	70	0.416	0.200	0.386	–
	Admitted farthest first	162	70	0.468	0.000	0.000	0.200
	Admitted coverage greedy	162	70	0.506	0.114	0.243	0.108
	Repair pool + Q	162	70	0.506	0.000	0.000	0.205
page-blocks0	Raw labeled random	195	49	0.905	0.061	0.559	0.390
	Schema random	136	49	0.889	0.069	0.559	0.477
	Admitted random	119	49	0.914	0.082	0.702	0.535
	Structure + Q selection	136	49	0.934	0.000	0.286	0.202
	Structure/local + Q	120	49	0.932	0.000	0.327	0.446
	Structure/global + Q	120	49	0.934	0.000	0.327	0.448
	Structure/local/global + Q	119	49	0.932	0.000	0.327	0.446
	Admitted source order	119	49	0.914	0.265	0.592	–
	Admitted farthest first	119	49	0.932	0.000	0.000	0.550
	Admitted coverage greedy	119	49	0.932	0.163	0.449	0.627
	Repair pool + Q	168	49	0.937	0.000	0.102	0.342

TABLE III

HISTORICAL EXTERNAL TRAINING BASELINES. ROS AND SMOTE/NC APPEND B MINORITY ROWS. LR: LOGISTIC REGRESSION; RF: RANDOM FOREST. THESE DIFFER FROM CANDIDATE CURATION.

Dataset / method	Model	$+n$	F1	AP	BA	MCC
Abalone19 / Real	LR	0	0.000	0.026	0.500	0.000
Abalone19 / Real	RF	0	0.000	0.023	0.500	0.000
Abalone19 / Weighted LR	LR	0	0.054	0.038	0.873	0.144
Abalone19 / ROS+b	LR	56	0.000	0.031	0.500	0.000
Abalone19 / ROS+b	RF	56	0.000	0.031	0.500	0.000
Abalone19 / SMOTENC+b	LR	56	0.000	0.014	0.499	-0.003
Abalone19 / SMOTENC+b	RF	56	0.000	0.020	0.499	-0.003
Ecoli4 / Real	LR	0	1.000	1.000	1.000	1.000
Ecoli4 / Real	RF	0	0.667	1.000	0.750	0.696
Ecoli4 / Weighted LR	LR	0	0.800	1.000	0.984	0.803
Ecoli4 / ROS+b	LR	70	1.000	1.000	1.000	1.000
Ecoli4 / ROS+b	RF	70	0.857	1.000	0.875	0.859
Ecoli4 / SMOTE+b	LR	70	1.000	1.000	1.000	1.000
Ecoli4 / SMOTE+b	RF	70	0.857	1.000	0.875	0.859
page-blocks0 / Real	LR	0	0.715	0.746	0.802	0.699
page-blocks0 / Real	RF	0	0.869	0.921	0.911	0.856
page-blocks0 / Weighted LR	LR	0	0.721	0.768	0.903	0.696
page-blocks0 / ROS+b	LR	49	0.728	0.746	0.811	0.712
page-blocks0 / ROS+b	RF	49	0.864	0.920	0.906	0.851
page-blocks0 / SMOTE+b	LR	49	0.728	0.758	0.811	0.712
page-blocks0 / SMOTE+b	RF	49	0.869	0.916	0.911	0.856

limit the rest. Ecoli4 logistic TRTR predicts its four positive and 63 negative TEST records perfectly, giving F1 interval $[1, 1]$ but a Wilson positive-recall interval of about $[.510, 1]$, and Abalone19 has six TEST positives; resampling adds no

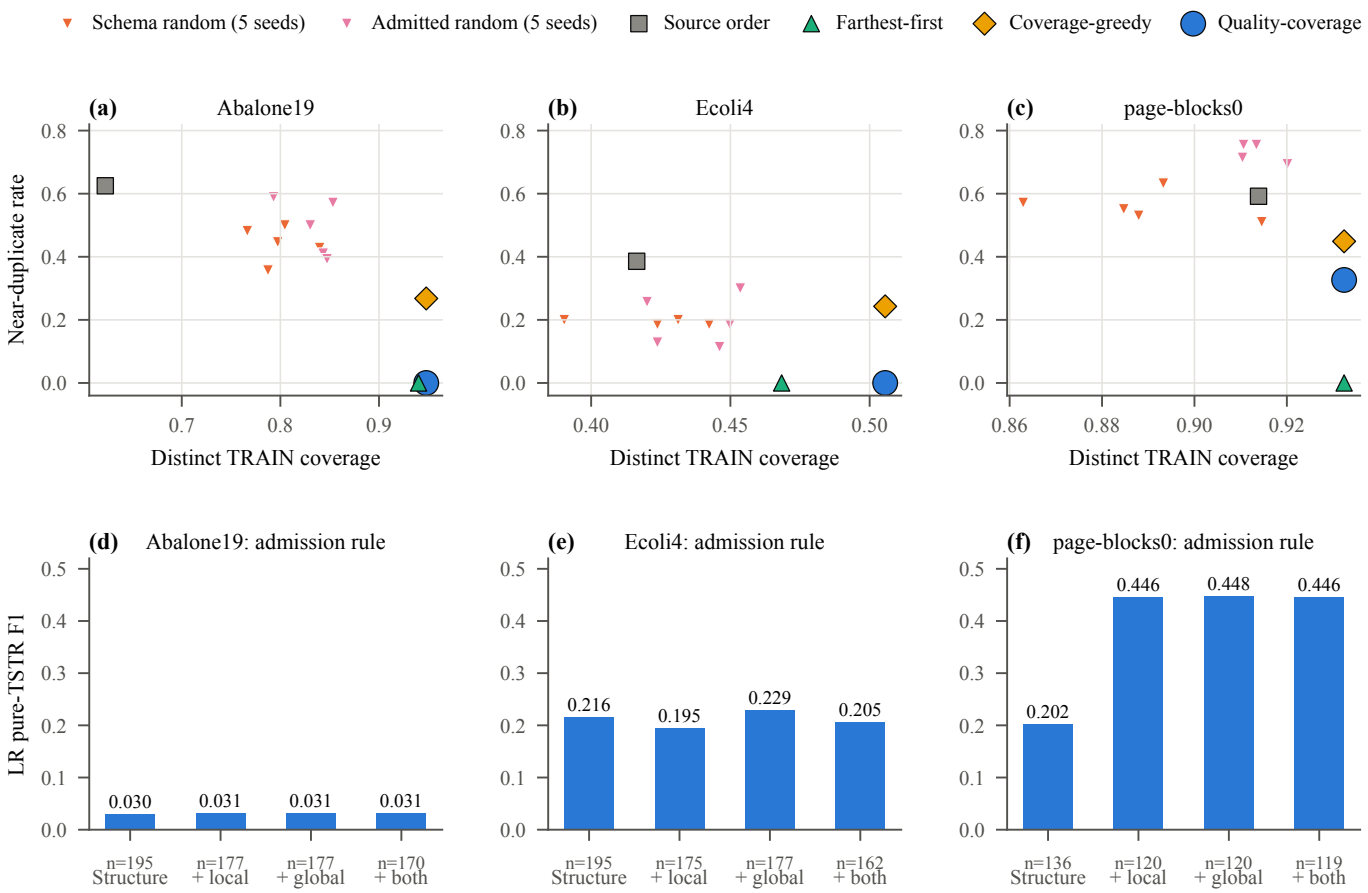


Fig. 4. Historical matched-budget construction. (a–c) Distinct-feature TRAIN coverage against near-duplicate rate; random controls show all five seeds, and deterministic selectors share the admitted pool (on Abalone19, farthest-first and quality-coverage nearly coincide). (d–f) Logistic pure-TSTR F1 after quality-coverage selection under each admission rule; n is the eligible pool.

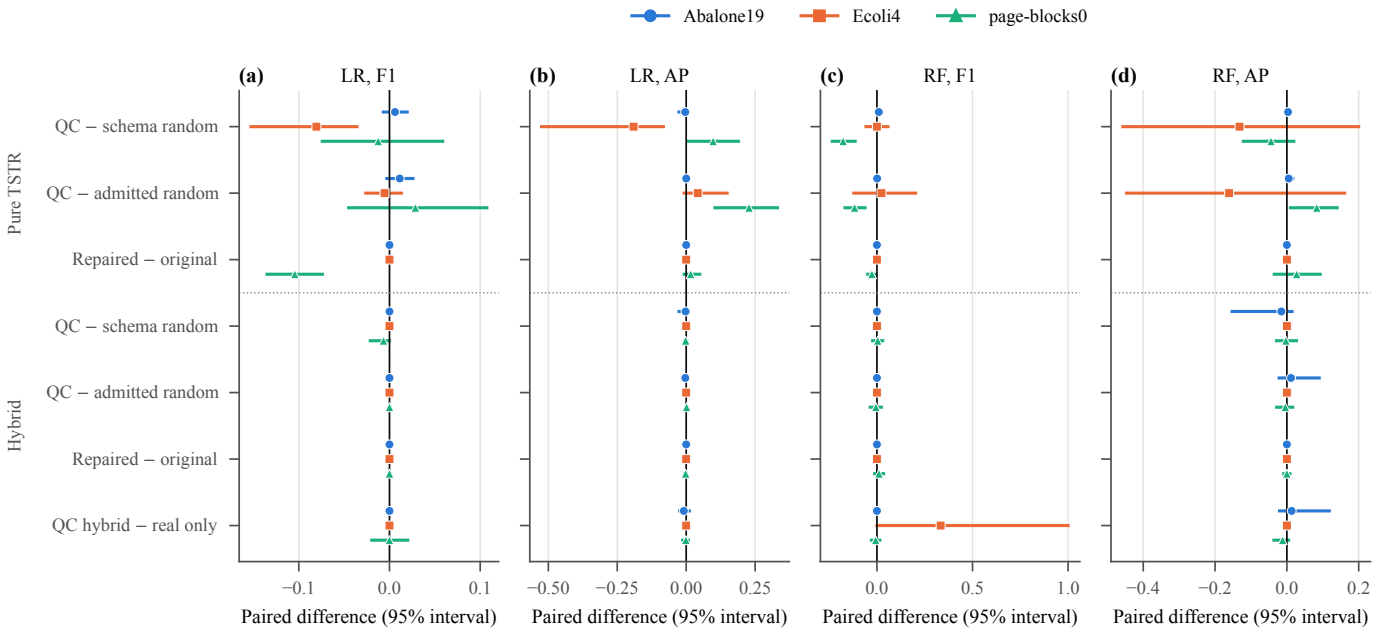


Fig. 5. Conditional paired 95% intervals for the primary historical pairs (2,000 stratified resamples of saved predictions); markers are observed differences. Intervals condition on the fitted models and the reused historical TEST set; endpoint intervals are in Appendix Table VI.

independent positive observations.

All 15 split-specific runs complete (Fig. 6). Ecoli4 logistic F1 is $.8476 \pm .1186$ for real-only training and $.7581 \pm .1659$ for quality-curated augmentation, the same mean as admitted random. Abalone19 F1 stays zero under real-only training and every augmentation method, while balanced logistic loss reaches $.0506 \pm .0095$. On page-blocks0, real-only F1 is $.7284 \pm .0256$, quality-curated augmentation $.7383 \pm .0283$ and admitted random $.7409 \pm .0270$. Augmentation brings no uniform benefit and does not beat simple curation.

F. Exploratory addendum

These experiments were declared after the main study, in settings files committed before any score was computed, and each ran once. They address four review gaps: few datasets, the half-budget minority quota, no equal-budget B2 comparison and no published baselines. Their results stay out of the preceding tables, and all tests are exploratory, without multiplicity adjustment.

Six KEEL datasets join the original three (Fig. 2; Appendix Table IX) under the RQ6 fresh-split design, with SMOTE or SMOTENC pools of size $4b$ and $b = \min(70, \max(30, \mathrm{round}(0.5\, n_{\min})))$; winequality-red-4 is unavailable because its source archive is empty. vehicle0 declares all 18 features as integers, unrounded SMOTE violates this in all 1,400 candidates, and bounded repair recovers six because nearly every record needs more than three edits. No curator can run on vehicle0, the first fresh-split case where structural checks change the outcome.

On the eight datasets with usable pools, quality-coverage does not beat random selection from the same pool (Fig. 7c; Appendix Table X). Median dataset-level F1 differences are $+.001$ (logistic regression) and $+.013$ (random forest); over 40 paired splits per classifier, wins, ties and losses are 11/22/7 and 15/15/10, and Wilcoxon p-values for F1 and AP range from .19 to 1.00. The geometric effect replicates on all five new datasets with usable pools (Fig. 7a,b): on abalone9-18, coverage is .677 versus .461 and near-duplication .000 versus .327. Picking the best augmentation per dataset can appear to beat real-only training, but uses the same TEST splits and is optimistic by construction.

On the historical split (Table IV), a minority quota of $\lfloor b \times$ TRAIN minority fraction$\rfloor$ (0, 4 and 5 instead of 28, 35 and 24) leaves coverage unchanged. Pure TSTR F1 rises on page-blocks0 (.446 to .593 for logistic regression, .342 to .631 for random forest) as near-duplication falls from .327 to zero, but falls on Ecoli4 (.296 to .250, random forest) and slightly on Abalone19, and hybrid F1 moves by at most .011, so the half-budget quota shapes synthetic-only results in both directions but does not mainly determine hybrid utility. At equal budget, quality-coverage covers more TRAIN support than random B2-accepted, structurally complete subsets (.948, .506, .932 versus .853, .419, .905), but F1 has no consistent winner: B2 subsets lead on page-blocks0 random-forest TSTR (.438 versus .342) and trail on Ecoli4 (.271 versus .296).

The second addendum adds two curators that re-implement published principles, not the authors' code: a CLLM-style learning-dynamics filter [5] that keeps low-aleatoric, high-confidence candidates under a gradient-boosting model fitted on real TRAIN, and a discriminator-realism filter after RUBRIC's realism component [8] and sample-level auditing [9] that keeps the candidates a random forest finds hardest to separate from real minority rows. All 45 rebuilt pools match their recorded fingerprints, and the 930 overlapping rows reproduce exactly. Among eight curators (Fig. 8a; Appendix Table XI), quality-coverage ranks best only for logistic-regression F1 (3.75 of 8), the CLLM-style filter for logistic-regression AP (2.44) and the discriminator for random-forest AP (2.88); Friedman tests do not separate the curators ($p = .21$ to .65).

Four imbalanced-learn oversamplers [26], SMOTE-ENN, SMOTE-Tomek [24], Borderline-SMOTE [27] and ADASYN [28], run with full balancing and with b added rows (Appendix Table XII). Full balancing helps random forest most on the hardest datasets (SMOTE-Tomek F1 .433 on yeast4 against .268 for quality-coverage; Fig. 8c) but often lowers logistic-regression F1, to .63–.68 on yeast3 against .78. Borderline-SMOTE and ADASYN cannot handle nominal features, the SMOTE-ENN and SMOTE-Tomek cleaning step fails on nominal columns, and ADASYN with b added rows generates nothing on page-blocks0, so the complete-case comparison covers five datasets. There, full-balancing SMOTE-Tomek and ADASYN have the best random-forest F1 ranks (3.70 and 4.50 against 7.60 for quality-coverage; Friedman $p = .031$; Fig. 8b, Appendix Table XIII), no method dominates the other three classifier–metric combinations, and no Wilcoxon comparison with quality-coverage reaches $p < .05$ (smallest .06). As context only, our real-only and SMOTE AUROC values exceed those of Newaz and Haq [29] on six shared datasets under 10-fold cross-validation, and the SMOTE effect's sign agrees in five of twelve dataset–classifier pairs.

On nine datasets, the addenda agree with the main study: quality-coverage reliably improves the coverage and redundancy of the curated set but not classifier utility over simple selection or published oversampling.

VI. Discussion

The strongest empirical contribution is a measurable separation of compliance, support, geometry and utility. Structural defects are frequent enough in this historical stream to motivate explicit checks, and narrow repair corrects an eligible subset without fabricating labels. Coverage-oriented construction reduces redundancy, but its classifier consequences are mixed, so geometric evidence should accompany direct task evaluation. Admission ablations measure the effect of changing candidate availability under fixed downstream rules, not a score's isolated utility effect, and repair changes both pool and selection, so its utility decrease cannot be assigned to rounding one field. Source order can exclude the minority entirely, which makes unavailable TSTR a meaningful result rather than a reason to add ineligible data. Simple alternatives are informative: farthest-first eliminates duplicates, random selection stays competitive

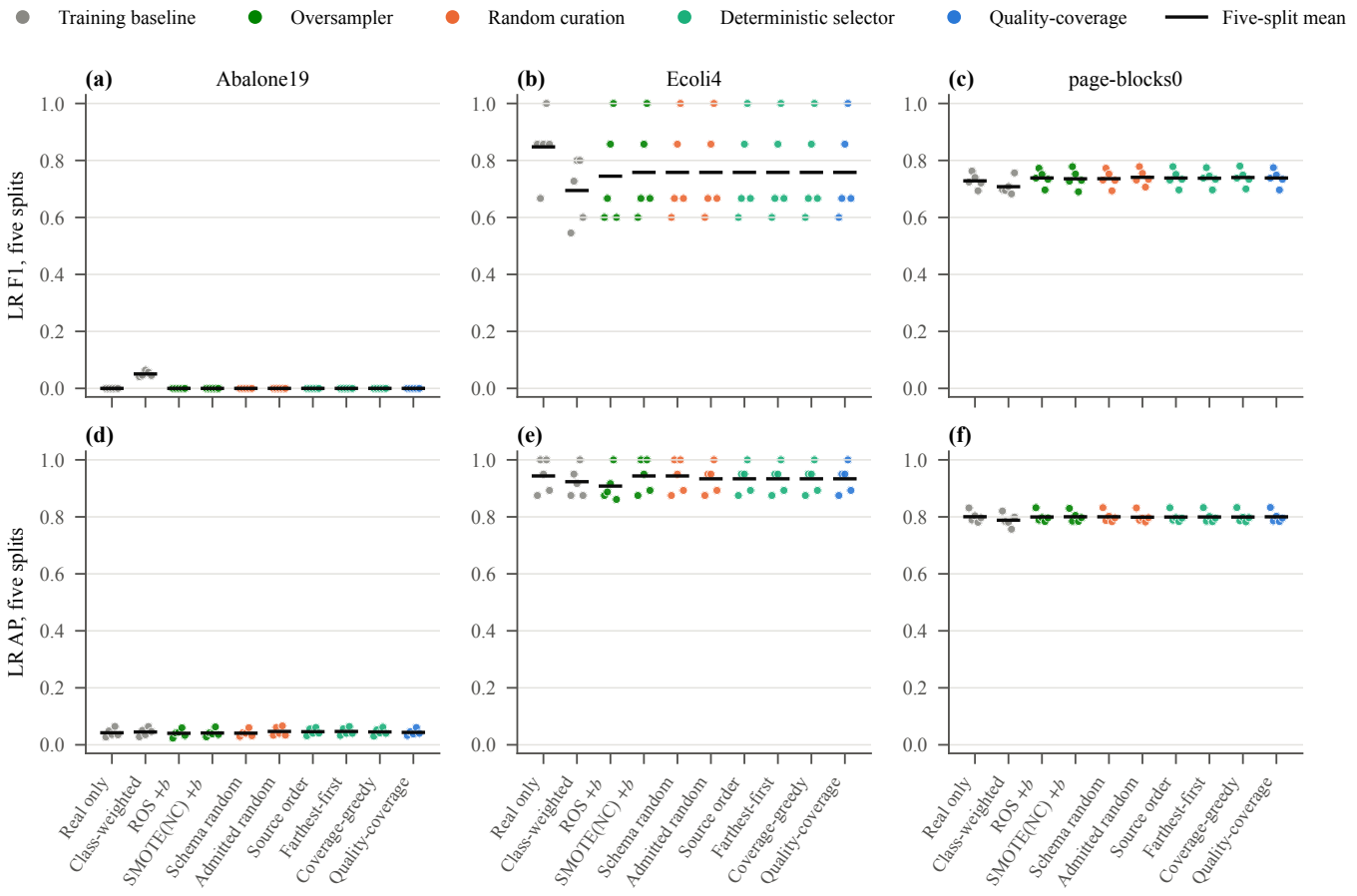


Fig. 6. Offline replication on five fresh stratified splits: logistic F1 (a–c) and AP (d–f) per split with five-split means. Curators use the hybrid regime; real-only and class-weighted training use TRTR; the page-blocks0 pool generator is ROS. Summary statistics are in Appendix Table VIII.

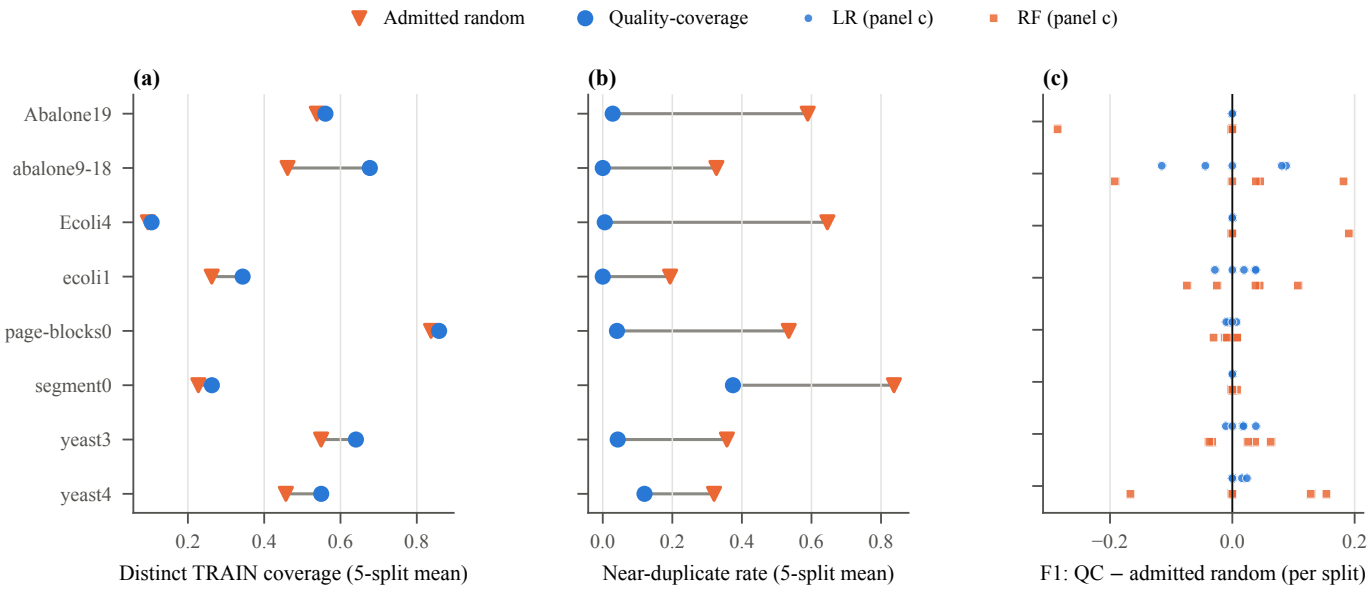


Fig. 7. Exploratory addendum on fresh splits. (a) Distinct TRAIN coverage and (b) near-duplicate rate of quality-coverage and admitted random selection from the same pool (five-split means). (c) Per-split F1 difference, quality-coverage minus admitted random, for logistic regression (LR) and random forest (RF). Main-study datasets use the RQ6 runs; vehicle0 has no usable pool.

TABLE IV

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. ADDENDUM V1 HISTORICAL-SPLIT DESIGNS AT FIXED BUDGETS: CANONICAL QUALITY-COVERAGE (QC, QUOTA $0.5b$), QC WITH THE REAL-RATIO QUOTA (E2), ADMITTED RANDOM (MEAN OF FIVE SEEDS) AND RANDOM SUBSETS OF THE B2-ACCEPTED POOL (E3, MEAN OF FIVE SEEDS).

Dataset	Design	Quota	Cov.	Near dup.	LR TSTR F1	LR Hyb. F1	RF TSTR F1	RF Hyb. F1	LR Hyb. AP
Abalone19	QC	28	.948	.000	.031	.000	.030	.000	.017
Abalone19	QC real ratio (E2)	0	.948	.000	.029	.000	.027	.000	.018
Abalone19	Adm. random	–	.833	.493	.022	.000	.030	.000	.021
Abalone19	B2 random (E3)	–	.853	.232	.026	.000	.030	.000	.015
Ecoli4	QC	35	.506	.000	.205	1.000	.296	1.000	1.000
Ecoli4	QC real ratio (E2)	4	.506	.000	.190	1.000	.250	1.000	1.000
Ecoli4	Adm. random	–	.439	.197	.217	.933	.231	1.000	1.000
Ecoli4	B2 random (E3)	–	.419	.049	.213	.933	.271	1.000	1.000
page-blocks0	QC	24	.932	.327	.446	.715	.342	.863	.744
page-blocks0	QC real ratio (E2)	5	.932	.000	.593	.722	.631	.873	.747
page-blocks0	Adm. random	–	.914	.702	.535	.715	.475	.869	.743
page-blocks0	B2 random (E3)	–	.905	.506	.421	.715	.438	.867	.744

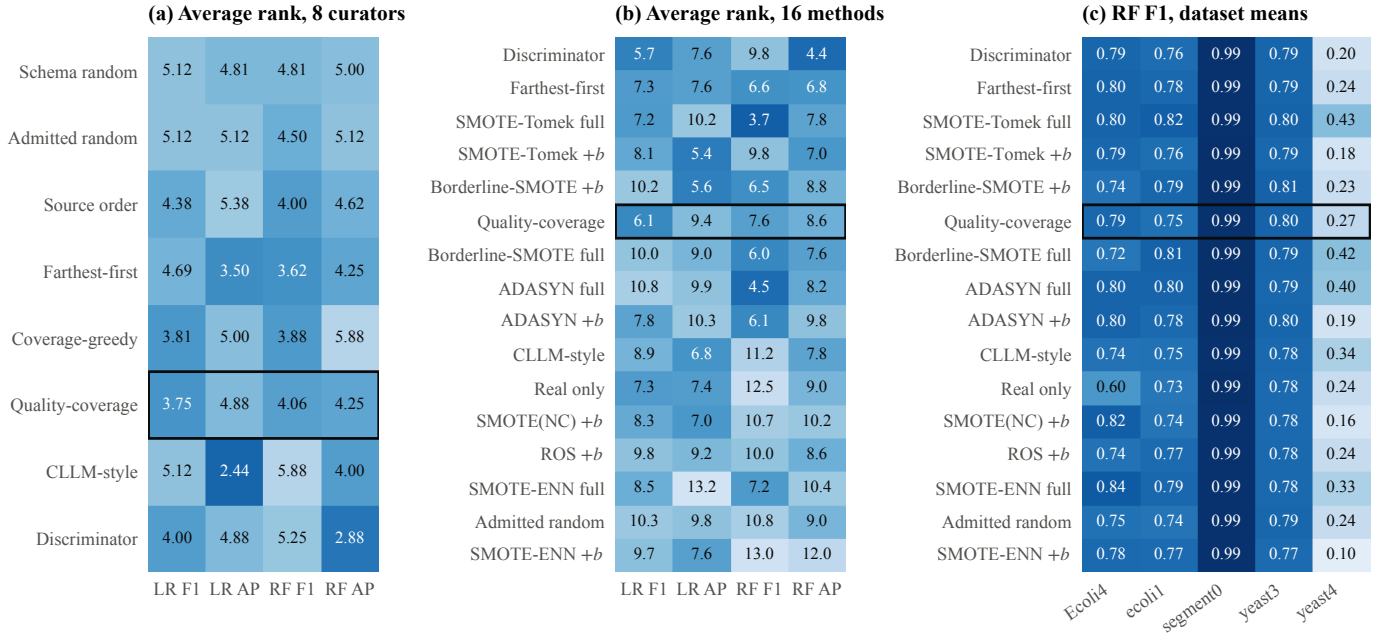


Fig. 8. Exploratory benchmark. (a) Average rank of the eight curators on the eight datasets with usable pools. (b) Average rank of all sixteen methods on the five complete-case datasets, sorted by mean rank. (c) Random-forest F1 dataset means for the same methods. Lower rank is better; the outlined row is quality-coverage. Full tables are in the appendix.

offline, and balanced logistic loss gives nonzero Abalone19 F1 where fixed-budget augmentation does not. The contribution is auditable curation with explicit tradeoffs, not universally better classifier training.

VII. LIMITATIONS

Data and splits. The historical study uses three datasets, a previously evaluated TEST set with recorded TRAIN/TEST overlap, and a pool generated from full original TRAIN, which prevents independent split claims. The fresh study rebuilds every fitted object, but its holdouts overlap, its generators differ from the LLM stream, ROS adds no new support and minority-only pools preclude pure TSTR. Offline findings, including the addendum, do not replicate LLM generation.

Inference. Ecoli4 and Abalone19 have four and six historical TEST positives, and abalone9-18 and yeast4 have eight and ten per split. Stratified intervals condition on observed records and fixed models, quantify neither generation nor refitting uncertainty, and cannot discover unseen positive modes; excluding undefined MCC margins can change the conditional target. All comparisons are exploratory, with no multiplicity-adjusted significance or state-of-the-art claim.

Validity. Source declarations are a limited contract: physical constraints, target meaning and cross-field semantics stay

unresolved where sources cannot justify them. Without a human annotation experiment, complete semantic validity is not independently established; LLM-proposed labels from an earlier phase for the 120-record audit sample (27 invalid, all structural; 93 indeterminate; none confirmed valid) are not ground truth and are unused. Evidence is uncalibrated, provisional admission is a selection policy rather than verified validity, distances depend on scaling and covariance assumptions, and coverage establishes neither predictive accuracy nor privacy.

Baselines and addendum. The utility evidence does not favour the proposed selector. Equal-budget B1 and B3 comparisons are unreported (B3 would need new judge calls), and the learned curators re-implement principles of CLLM and RUBRIC's realism component without CLLM's LLM generation or RUBRIC's boundary-utility term. The addendum was designed after the main results and review were known, so it tests declared questions rather than replicating independently. Several datasets share KEEL sources (yeast3/yeast4, Ecoli4/ecoli1, Abalone19/abalone9-18), so dataset-level tests overstate the effective sample. Re-scoring the canonical comparators on another platform changes some metrics by up to .0097, and the published AUROC comparison uses a different protocol. Structural checks and repair are exercised mainly on the historical LLM stream; on vehicle0, the only fresh pool with defects, the three-edit bound recovers six of 1,400 candidates, so the repair policy does not suit interpolation output.

Future work. Open directions are independent datasets with more minority TEST records beyond the KEEL derivatives used here, a second LLM generation campaign under fresh splits, structural rules and repair bounds designed for interpolation output, equal-budget B1 and B3 comparisons with full learned-curation implementations, and blinded human validity labels for the audit sample.

VIII. Conclusion

RC-QASyn makes provenance, structural compliance, statistical support, fixed-budget construction and bounded repair inspectable within the existing six-phase system. The historical audit identifies 75 structural rejects, repairs all 54 eligible records and measures the resulting changes in coverage and redundancy. Paired uncertainty and fresh train-only experiments strengthen the evidence while revealing limited and sometimes negative utility effects, and an exploratory addendum on nine datasets reproduces the geometric gain without a utility advantage over random selection, re-implemented learned curators or published oversamplers. The contribution is an empirically bounded, reproducible account of curation tradeoffs.

References

[1] L. Xu, M. Skoularidou, A. Cuesta-Infante, and K. Veeramachaneni, "Modeling tabular data using conditional GAN," in *Advances in Neural Information Processing Systems*, vol. 32, 2019. [Online]. Available: https://proceedings.neurips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html

[2] A. Kotelnikov, D. Baranchuk, I. Rubachev, and A. Babenko, "TabDDPM: Modelling tabular data with diffusion models," in *Proceedings of the 40th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 202. PMLR, 2023, pp. 17 564–17 579. [Online]. Available: https://proceedings.mlr.press/v202/kotelnikov23a.html

[3] V. Borisov, K. Seßler, T. Leemann, M. Pawelczyk, and G. Kasneci, "Language models are realistic tabular data generators," *arXiv preprint arXiv:2210.06280*, 2023. [Online]. Available: https://arxiv.org/abs/2210.06280

[4] J. Kim, T. Kim, and J. Choo, "EPIC: Effective prompting for imbalanced-class data synthesis in tabular data classification via large language models," in *Advances in Neural Information Processing Systems*, vol. 37, 2024.

[5] N. Seedat, N. Huynh, B. van Breugel, and M. van der Schaar, "Curated LLM: Synergy of LLMs and data curation for tabular augmentation in low-data regimes," in *Proceedings of the 41st International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 235, 2024, pp. 44 060–44 092.

[6] M. Jiang, K. Chen, Y. Wang, Y. Zhao, Z. Dai, Y. Liu, Z. Li, Q. Zhang, H. Nie, H. Zhu, S. Liu, G. Ye, and H. Chai, "Limited reference, reliable generation: A two-component framework for tabular data generation in low-data regimes," arXiv preprint arXiv:2509.09960, 2025.

[7] C. Zhang, R. F. Lin, R. Bao, and S. Huang, "Team, then trim: An assembly-line LLM framework for high-quality tabular data generation," arXiv preprint arXiv:2602.04785, 2026.

[8] Y. Yu, D. Liu, S. Wang, W. Zhao, E. Jiang, C. Liu, J. Yu, H. Pan, R. Borovica-Gajic, and B. Lengerich, "RUBRIC: Realism–utility balanced ranking for imbalanced classification," arXiv preprint arXiv:2607.09816, 2026.

[9] A. Alaa, B. V. Breugel, E. S. Saveliev, and M. van der Schaar, "How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models," in *Proceedings of the 39th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 162. PMLR, 2022, pp. 290–306. [Online]. Available: https://proceedings.mlr.press/v162/alaa22a.html

[10] M. F. Naeem, S. J. Oh, Y. Uh, Y. Choi, and J. Yoo, "Reliable fidelity and diversity metrics for generative models," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119, 2020, pp. 7176–7185.

[11] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher, "An analysis of approximations for maximizing submodular set functions—i," *Mathematical Programming*, vol. 14, pp. 265–294, 1978. [Online]. Available: https://link.springer.com/article/10.1007/BF01588971

[12] O. Sener and S. Savarese, "Active learning for convolutional neural networks: A core-set approach," in *International Conference on Learning Representations*, 2018. [Online]. Available: https://openreview.net/forum?id=H1aIuk-RW

[13] S. Tavakkol, M. Springer, M. Bateni, N. Bulut, V. Cohen-Addad, and M. Hajiaghayi, "Less is more: Adaptive coverage for synthetic training data," arXiv preprint arXiv:2504.14508, 2025.

[14] W. M. Stevenson, A. Gerdes, L. Galke Poech, and A. D. Lautrup, "Curating the collapse: How sample-level auditing accelerates model collapse in synthesis of tabular data," in *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society*, vol. 9, no. 4, 2026, pp. 2450–2461.

[15] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proceedings of the 34th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 70. PMLR, 2017, pp. 1321–1330. [Online]. Available: https://proceedings.mlr.press/v70/guo17a.html

[16] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979. [Online]. Available: https://doi.org/10.1214/aos/1176344552

[17] L. D. Brown, T. T. Cai, and A. DasGupta, "Interval estimation for a binomial proportion," *Statistical Science*, vol. 16, no. 2, pp. 101–133, 2001. [Online]. Available: https://doi.org/10.1214/ss/1009213286

[18] L. Long, R. Wang, R. Xiao, J. Zhao, X. Ding, G. Chen, and H. Wang, "On LLMs-driven synthetic data generation, curation, and evaluation: A survey," in *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, 2024, pp. 11 065–11 082. [Online]. Available: https://aclanthology.org/2024.findings-acl.658/

[19] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and

I. Stoica, "Judging LLM-as-a-judge with MT-bench and chatbot arena," in *Advances in Neural Information Processing Systems*, vol. 36, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/91f18a1287b398d378ef22505bf41832-Abstract-Datasets_and_Benchmarks.html

[20] E. Breck, N. Polyzotis, S. Roy, S. E. Whang, and M. Zinkevich, "Data validation for machine learning," in *Proceedings of Machine Learning and Systems (MLSys)*, 2019.

[21] M. Stoian, S. Dyrmishi, M. Cordy, T. Lukasiewicz, and E. Giunchiglia, "How realistic is your synthetic data? constraining deep generative models for tabular data," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/887932131fddf943e8fe3310b62c0147-Abstract-Conference.html

[22] J. Zhao, M. Guan, and D. Liu, "Constraint-aware synthetic tabular data generation via inter-column constraint discovery with LLM agents," arXiv preprint arXiv:2608.15109, 2026.

[23] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002. [Online]. Available: https://www.jair.org/index.php/jair/article/view/10302

[24] G. E. A. P. A. Batista, R. C. Prati, and M. C. Monard, "A study of the behavior of several methods for balancing machine learning training data," *ACM SIGKDD Explorations Newsletter*, vol. 6, no. 1, pp. 20–29, 2004.

[25] J. A. Sáez, J. Luengo, J. Stefanowski, and F. Herrera, "SMOTE-IPF: Addressing the noisy and borderline examples problem in imbalanced classification by a re-sampling method with filtering," *Information Sciences*, vol. 291, pp. 184–203, 2015.

[26] G. Lemaître, F. Nogueira, and C. K. Aridas, "Imbalanced-learn: A python toolbox to tackle the curse of imbalanced datasets in machine learning," *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017. [Online]. Available: https://www.jmlr.org/papers/v18/16-365.html

[27] H. Han, W.-Y. Wang, and B.-H. Mao, "Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning," in *Advances in Intelligent Computing*, ser. Lecture Notes in Computer Science. Berlin, Heidelberg: Springer, 2005, pp. 878–887.

[28] H. He, Y. Bai, E. A. Garcia, and S. Li, "ADASYN: Adaptive synthetic sampling approach for imbalanced learning," in *2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)*. IEEE, 2008, pp. 1322–1328.

[29] A. Newaz and F. S. Haq, "A novel hybrid sampling framework for imbalanced learning," arXiv preprint arXiv:2208.09619, 2022.

[30] J. Alcalá-Fdez, A. Fernández, J. Luengo, J. Derrac, S. García, L. Sánchez, and F. Herrera, "KEEL data-mining software tool: Data set repository, integration of algorithms and experimental analysis framework," *Journal of Multiple-Valued Logic and Soft Computing*, vol. 17, no. 2–3, pp. 255–287, 2011. [Online]. Available: https://sci2s.ugr.es/keel/imbalanced.php

APPENDIX A
CLASS DENOMINATORS AND CONFUSION COUNTS

APPENDIX B
CONDITIONAL INTERVALS AND FRESH-SPLIT SUMMARIES

All primary pairs for both classifiers and available regimes are in `paired-differences.json`. Undefined counts remain per endpoint. `confusions.csv` retains confusion counts and Wilson recall bounds for every successful result.

APPENDIX C
EXPLORATORY ADDENDUM TABLES

APPENDIX D
EVIDENCE MAP AND REPRODUCTION

The protocol is `docs/empirical-v2/protocol.md`; machine settings are in `configs/empirical/protocol-v1.json`. Canonical evidence is `artifacts/reports/empirical-v2/run-v2`, and the report includes consolidated ablations, split-wise paired differences and confusions. A development run is explicitly superseded for an interval-metadata naming defect; its scores never selected configurations.

With this revision's `src` on PYTHONPATH, run `scripts/run_empirical_study.py` with a fresh output directory and the frozen checkout as `-source-root`. `scripts/report_empirical_study.py` reads saved evidence without refitting, and `scripts/build_ieee_figures.py` draws this paper's figures from the same saved evidence and the addendum reports. Exact commands, pinned dependencies and verification are in `docs/empirical-v2/handoff.md`.

The manifest binds revisions, code hashes, protocol identity, dependencies, runtime and output hashes. Population records retain exact TRAIN/TEST identities, source and dataset fingerprints, class counts and overlap diagnostics. Generator files retain fitted TRAIN identities, scaler parameters, seeds and generated payloads, and existing M2/M7 use fresh split-specific populations. Predictions preserve actual and predicted classes, positive scores and row identity, and compressed index arrays preserve paired resamples.

The cell ledger `artifacts/reports/empirical-v2/report-v3/table-evidence.json` links rendered values to input fields, row fingerprints and aggregation rules. Original M2, M4 and M6/M7 sensitivities remain unchanged. Individual seeds, every split, both classifiers, unavailable cases and budget checks accompany the paper. Full regression and static checks, replay and historical integrity are reported separately; replay is not another independent split.

TABLE V

HISTORICAL PRIMARY PREDICTIONS: CLASS DENOMINATORS AND CONFUSION COUNTS. Q AND Q REPAIR USE PURE TSTR; REAL USES TRTR.

Dataset / method	Model	n_+	n_-	TP	FP	TN	FN
Abalone19 / Real	LR	6	828	0	0	828	6
Abalone19 / Real	RF	6	828	0	0	828	6
Abalone19 / Q	LR	6	828	6	373	455	0
Abalone19 / Q	RF	6	828	5	319	509	1
Abalone19 / Q repair	LR	6	828	6	373	455	0
Abalone19 / Q repair	RF	6	828	5	319	509	1
Ecoli4 / Real	LR	4	63	4	0	63	0
Ecoli4 / Real	RF	4	63	2	0	63	2
Ecoli4 / Q	LR	4	63	4	31	32	0
Ecoli4 / Q	RF	4	63	4	19	44	0
Ecoli4 / Q repair	LR	4	63	4	31	32	0
Ecoli4 / Q repair	RF	4	63	4	19	44	0
page-blocks0 / Real	LR	112	983	69	12	971	43
page-blocks0 / Real	RF	112	983	93	9	974	19
page-blocks0 / Q	LR	112	983	75	149	834	37
page-blocks0 / Q	RF	112	983	85	300	683	27
page-blocks0 / Q repair	LR	112	983	78	266	717	34
page-blocks0 / Q repair	RF	112	983	84	336	647	28

TABLE VI

CONDITIONAL 95% BOOTSTRAP INTERVALS (2,000 STRATIFIED PAIRED RESAMPLES). Q: QUALITY-COVERAGE PURE TSTR; REAL: TRTR. DEGENERATE MCC INTERVALS ARE UNAVAILABLE; ALL VALID COUNTS ARE SAVED.

Dataset / method	Model	F1 [CI]	AP [CI]	BA [CI]	MCC [CI]
Abalone19 / Real	LR	0.000 [0.000, 0.000]	0.026 [0.022, 0.044]	0.500 [0.500, 0.500]	0.000 [–, –]
Abalone19 / Real	RF	0.000 [0.000, 0.000]	0.023 [0.012, 0.047]	0.500 [0.500, 0.500]	0.000 [–, –]
Abalone19 / Q	LR	0.031 [0.029, 0.034]	0.019 [0.012, 0.068]	0.775 [0.758, 0.791]	0.093 [0.087, 0.100]
Abalone19 / Q	RF	0.030 [0.018, 0.039]	0.021 [0.011, 0.046]	0.724 [0.553, 0.820]	0.078 [0.018, 0.112]
Abalone19 / Q repair	LR	0.031 [0.029, 0.034]	0.019 [0.012, 0.068]	0.775 [0.758, 0.791]	0.093 [0.087, 0.100]
Abalone19 / Q repair	RF	0.030 [0.018, 0.039]	0.021 [0.011, 0.046]	0.724 [0.553, 0.820]	0.078 [0.018, 0.112]
Ecoli4 / Real	LR	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]
Ecoli4 / Real	RF	0.667 [0.000, 1.000]	1.000 [1.000, 1.000]	0.750 [0.500, 1.000]	0.696 [0.489, 1.000]
Ecoli4 / Q	LR	0.205 [0.170, 0.258]	0.135 [0.109, 0.267]	0.754 [0.690, 0.817]	0.241 [0.188, 0.307]
Ecoli4 / Q	RF	0.296 [0.235, 0.400]	0.633 [0.348, 1.000]	0.849 [0.794, 0.905]	0.349 [0.280, 0.450]
Ecoli4 / Q repair	LR	0.205 [0.170, 0.258]	0.135 [0.109, 0.267]	0.754 [0.690, 0.817]	0.241 [0.188, 0.307]
Ecoli4 / Q repair	RF	0.296 [0.235, 0.400]	0.633 [0.348, 1.000]	0.849 [0.794, 0.905]	0.349 [0.280, 0.450]
page-blocks0 / Real	LR	0.715 [0.637, 0.783]	0.746 [0.663, 0.832]	0.802 [0.754, 0.845]	0.699 [0.619, 0.769]
page-blocks0 / Real	RF	0.869 [0.819, 0.914]	0.921 [0.868, 0.965]	0.911 [0.872, 0.944]	0.856 [0.802, 0.904]
page-blocks0 / Q	LR	0.446 [0.395, 0.500]	0.555 [0.458, 0.656]	0.759 [0.715, 0.804]	0.389 [0.325, 0.456]
page-blocks0 / Q	RF	0.342 [0.308, 0.377]	0.447 [0.368, 0.534]	0.727 [0.685, 0.768]	0.288 [0.234, 0.342]
page-blocks0 / Q repair	LR	0.342 [0.303, 0.383]	0.572 [0.482, 0.676]	0.713 [0.669, 0.758]	0.278 [0.222, 0.337]
page-blocks0 / Q repair	RF	0.316 [0.283, 0.350]	0.474 [0.391, 0.572]	0.704 [0.661, 0.748]	0.254 [0.200, 0.309]

TABLE VII

REPAIR-POOL MINUS ORIGINAL Q, LOGISTIC PURE TSTR: CONDITIONAL PAIRED DIFFERENCE INTERVALS. THESE MEASURE POOL EXPANSION AND RESELECTION TOGETHER.

Dataset	Metric	Lower	Upper	Valid	Undefined
Abalone19	f1	0.000	0.000	2000	0
Abalone19	average_precision	0.000	0.000	2000	0
Abalone19	balanced_accuracy	0.000	0.000	2000	0
Abalone19	mcc	0.000	0.000	2000	0
Ecoli4	f1	0.000	0.000	2000	0
Ecoli4	average_precision	0.000	0.000	2000	0
Ecoli4	balanced_accuracy	0.000	0.000	2000	0
Ecoli4	mcc	0.000	0.000	2000	0
page-blocks0	f1	-0.135	-0.074	2000	0
page-blocks0	average_precision	-0.007	0.049	2000	0
page-blocks0	balanced_accuracy	-0.067	-0.024	2000	0
page-blocks0	mcc	-0.146	-0.075	2000	0

TABLE VIII

OFFLINE REPLICATION: FIVE REPEATED STRATIFIED HOLDOUTS, LOGISTIC REGRESSION, MEAN ± SAMPLE SD. REAL AND WEIGHTED ARE TRTR; OTHER METHODS ARE HYBRID MINORITY AUGMENTATION. NO HISTORICAL CANDIDATES ARE REUSED.

Dataset / method	k	F1	AP	BA	MCC
Abalone19 / Weighted LR	5	0.051 ± 0.009	0.045 ± 0.014	0.779 ± 0.071	0.116 ± 0.028
Abalone19 / Real	5	0.000 ± 0.000	0.042 ± 0.015	0.500 ± 0.000	0.000 ± 0.000
Abalone19 / ROS+b	5	0.000 ± 0.000	0.040 ± 0.014	0.500 ± 0.001	-0.002 ± 0.002
Abalone19 / SMOTENC+b	5	0.000 ± 0.000	0.041 ± 0.013	0.500 ± 0.000	-0.002 ± 0.002
Abalone19 / Curated Q	5	0.000 ± 0.000	0.044 ± 0.011	0.500 ± 0.001	-0.002 ± 0.002
Abalone19 / Admitted random	5	0.000 ± 0.000	0.047 ± 0.016	0.500 ± 0.000	-0.001 ± 0.002
Abalone19 / Schema random	5	0.000 ± 0.000	0.041 ± 0.012	0.500 ± 0.000	-0.002 ± 0.002
Ecoli4 / Weighted LR	5	0.695 ± 0.117	0.923 ± 0.053	0.928 ± 0.074	0.692 ± 0.127
Ecoli4 / Real	5	0.848 ± 0.119	0.944 ± 0.058	0.875 ± 0.088	0.855 ± 0.108
Ecoli4 / ROS+b	5	0.745 ± 0.177	0.908 ± 0.055	0.865 ± 0.089	0.744 ± 0.182
Ecoli4 / SMOTE+b	5	0.758 ± 0.166	0.944 ± 0.058	0.867 ± 0.089	0.757 ± 0.170
Ecoli4 / Curated Q	5	0.758 ± 0.166	0.934 ± 0.050	0.867 ± 0.089	0.757 ± 0.170
Ecoli4 / Admitted random	5	0.758 ± 0.166	0.934 ± 0.050	0.867 ± 0.089	0.757 ± 0.170
Ecoli4 / Schema random	5	0.758 ± 0.166	0.944 ± 0.058	0.867 ± 0.089	0.757 ± 0.170
page-blocks0 / Weighted LR	5	0.708 ± 0.029	0.788 ± 0.024	0.913 ± 0.013	0.687 ± 0.031
page-blocks0 / Real	5	0.728 ± 0.026	0.800 ± 0.019	0.819 ± 0.025	0.709 ± 0.025
page-blocks0 / ROS+b	5	0.738 ± 0.028	0.800 ± 0.019	0.828 ± 0.028	0.718 ± 0.029
page-blocks0 / SMOTE+b	5	0.736 ± 0.033	0.800 ± 0.019	0.826 ± 0.031	0.715 ± 0.032
page-blocks0 / Curated Q	5	0.738 ± 0.028	0.800 ± 0.020	0.828 ± 0.029	0.718 ± 0.029
page-blocks0 / Admitted random	5	0.741 ± 0.027	0.799 ± 0.019	0.829 ± 0.030	0.721 ± 0.026
page-blocks0 / Schema random	5	0.736 ± 0.030	0.800 ± 0.019	0.826 ± 0.029	0.716 ± 0.029

TABLE IX

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. DATASETS. IR = MAJORITY/MINORITY; b = SELECTION BUDGET; TEST POSITIVES ARE PER SPLIT (FIVE STRATIFIED 80/20 SPLITS).

Dataset	Rows	Min.	IR	Nom.	b	TEST pos.	Pool gen.
Abalone19	4174	32	129.4	yes	56	6	SMOTENC
Ecoli4	336	20	15.8	no	70	4	SMOTE
page-blocks0	5472	559	8.8	no	49	112	RandomOverSampler
yeast3	1484	163	8.1	no	65	33	SMOTE
yeast4	1484	51	28.1	no	30	10	SMOTE
vehicle0	846	199	3.3	no	70	40	SMOTE
segment0	2308	329	6.0	no	70	66	SMOTE
ecoli1	336	77	3.4	no	31	15	SMOTE
abalone9-18	731	42	16.4	yes	30	8	SMOTENC
winequality-red-4	unavailable (empty source archive)						

TABLE X

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. ADDENDUM V1 ACROSS-DATASET SUMMARY: DATASET-LEVEL MEAN DIFFERENCE AND SPLIT WINS/TIES/LOSSES OVER FIVE FRESH SPLITS.

Comparison	Clf	Metric	Datasets $+/-/0$	Splits W/T/L	Median diff.
QC − adm. random	LR	AP	3/4/1	17/7/16	-0.001
QC − adm. random	LR	F1	4/1/3	11/22/7	+0.001
QC − adm. random	RF	AP	5/3/0	19/6/15	+0.002
QC − adm. random	RF	F1	6/2/0	15/15/10	+0.013
QC − real only	LR	AP	3/5/0	18/6/16	-0.000
QC − real only	LR	F1	5/1/2	15/15/10	+0.006
QC − real only	RF	AP	5/3/0	17/5/18	+0.002
QC − real only	RF	F1	5/2/1	18/11/11	+0.027
Best aug. − real only*	LR	AP	6/2/1	25/9/11	+0.003
Best aug. − real only*	LR	F1	6/1/2	18/17/10	+0.006
Best aug. − real only*	RF	AP	8/1/0	23/6/16	+0.021
Best aug. − real only*	RF	F1	7/2/0	28/9/8	+0.035

*Best method chosen on the same TEST splits; optimistic by construction.

TABLE XI

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. MEAN F1 AND AP OF THE EIGHT CURATORS (HYBRID TRAINING: REAL TRAIN PLUS b SELECTED CANDIDATES).

Dataset	Schema rand.	Adm. rand.	Source order	Farthest-first	Cov.-greedy	QC (ours)	CLLM-style†	Disc.-realism†
LR, mean F1 over five splits								
Abalone19	**.000**	**.000**	**.000**	**.000**	**.000**	**.000**	**.000**	**.000**
Ecoli4	.758	.758	.758	.758	.758	.758	**.826**	.758
page-blocks0	.736	**.741**	.738	.738	.740	.738	.736	.739
yeast3	.773	.769	.779	.773	**.782**	**.782**	.771	.779
yeast4	.152	.150	.154	**.203**	.158	.158	.180	.185
segment0	**.988**	**.988**	**.988**	**.988**	**.988**	**.988**	**.988**	**.988**
ecoli1	.760	.745	.749	.750	.762	.758	.742	**.762**
abalone9-18	.676	.677	**.684**	.670	.670	.679	.671	.659
LR, mean AP over five splits								
Abalone19	.041	.047	.046	.047	.045	.044	**.050**	.042
Ecoli4	**.944**	.934	.934	.934	.934	.934	**.944**	.937
page-blocks0	.800	.799	.799	.800	.799	.800	**.800**	.799
yeast3	.813	.809	.807	.809	.807	.808	**.813**	.813
yeast4	.311	.319	.311	.327	.306	.305	**.331**	.322
segment0	.977	.977	.977	.978	.977	**.978**	.974	.977
ecoli1	.861	.862	**.868**	.864	.867	.865	.864	.861
abalone9-18	.774	.780	.771	.780	.781	.775	**.783**	.780
RF, mean F1 over five splits								
Abalone19	.000	**.057**	.000	.000	.000	.000	.000	.000
Ecoli4	.786	.748	**.819**	.800	.762	.786	.743	.790
page-blocks0	.890	**.893**	.892	.891	.886	.885	.890	.885
yeast3	.775	.790	.787	.794	**.810**	.801	.781	.794
yeast4	.246	.245	.240	.238	.210	.268	**.335**	.199
segment0	.991	.989	.989	**.991**	**.991**	**.991**	.989	.989
ecoli1	.775	.736	.771	**.782**	.773	.754	.748	.762
abalone9-18	.348	.373	.402	.370	**.406**	.388	.313	.392
RF, mean AP over five splits								
Abalone19	.040	**.070**	.034	.030	.037	.033	.039	.041
Ecoli4	.884	.872	**.895**	.882	.870	.882	.811	.883
page-blocks0	.922	.922	.924	.925	.924	.926	**.929**	.924
yeast3	.859	.867	.855	.873	.858	.859	.875	**.875**
yeast4	.406	.379	.401	.415	.364	.407	.383	**.421**
segment0	1.000	1.000	**1.000**	1.000	1.000	1.000	1.000	1.000
ecoli1	.867	.862	.857	.860	.840	.820	.868	**.870**
abalone9-18	.480	.533	.511	.551	**.567**	.554	.497	.518

Bold: best in row. †Re-implementation of the published principle, not the authors' code. vehicle0 omitted: no structurally complete candidate.

TABLE XII

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. MEAN F1 AND AP OF REAL-ONLY TRAINING, CLASS WEIGHTING, EXISTING OVERSAMPLERS, FOUR PUBLISHED OVERSAMPLERS (IMBALANCED-LEARN 0.14.2; FULL BALANCING AND $+b$ ROWS) AND QUALITY-COVERAGE.

Dataset	Real only	CW-LR	ROS+b	SMOTE(NC)+b	SMOTE-ENN full	SMOTE-ENN +b	SMOTE-Tomek full	SMOTE-Tomek +b	B-SMOTE full	B-SMOTE +b	ADASYN full	ADASYN +b	QC (ours)
LR, mean F1 over five splits													
Abalone19	.000	**.051**	.000	.000	–	–	–	–	–	–	–	–	.000
Ecoli4	**.848**	.695	.745	.758	.711	.758	.745	.758	.700	.723	.688	.740	.758
page-blocks0	.728	.708	.738	.736	.693	**.761**	.699	.743	.665	.738	.628	–	.738
yeast3	.776	.660	.770	.778	.648	.779	.676	.776	.634	**.782**	.641	.776	.782
yeast4	.101	.235	.147	.154	.209	.099	.231	.200	**.282**	.201	.217	.222	.158
vehicle0	.927	.923	.931	.929	.892	.907	.922	.931	.919	.930	.918	**.934**	–
segment0	.988	.965	.988	.988	**.991**	.988	**.991**	.988	.931	.986	.938	.989	.988
ecoli1	.751	.746	.754	.749	.749	.737	.749	.745	**.760**	.730	.757	.744	.758
abalone9-18	.672	.471	.666	.636	–	–	–	–	–	–	–	–	**.679**
LR, mean AP over five splits													
Abalone19	.042	**.045**	.040	.041	–	–	–	–	–	–	–	–	.044
Ecoli4	.944	.923	.908	.944	.896	**.947**	.901	.940	.897	.922	.903	.903	.934
page-blocks0	**.800**	.788	.800	.800	.789	.794	.790	.800	.752	.800	.748	–	.800
yeast3	.813	.809	.807	.809	.805	.811	.807	**.814**	.795	.810	.801	.808	.808
yeast4	.313	.251	.308	.309	.242	**.338**	.259	.329	.312	.323	.254	.307	.305
vehicle0	.982	.976	.980	.981	.927	.921	**.984**	.981	.981	.983	.980	.981	–
segment0	.976	.981	.977	.977	.985	.973	.986	.977	**.997**	.983	.996	.983	.978
ecoli1	.866	.870	**.874**	.870	.842	.842	.867	.867	.870	.872	.868	.865	.865
abalone9-18	.771	**.777**	.772	.766	–	–	–	–	–	–	–	–	.775
RF, mean F1 over five splits													
Abalone19	**.000**	–	**.000**	**.000**	–	–	–	–	–	–	–	–	**.000**
Ecoli4	.600	–	.743	.819	**.836**	.776	.800	.786	.724	.743	.798	.798	.786
page-blocks0	.893	–	.885	**.901**	.830	.856	.865	.895	.864	.896	.850	–	.885
yeast3	.775	–	.783	.777	.778	.771	.804	.786	.792	**.807**	.788	.795	.801
yeast4	.236	–	.243	.165	.329	.103	**.433**	.185	.425	.234	.405	.188	.268
vehicle0	.928	–	.927	.927	.877	.884	.920	.928	.924	**.931**	.907	.928	–
segment0	.992	–	.991	.991	.989	.989	.991	.992	.992	.994	.994	**.994**	.991
ecoli1	.726	–	.771	.738	.787	.772	**.820**	.760	.806	.789	.797	.776	.754
abalone9-18	.347	–	.373	.374	–	–	–	–	–	–	–	–	**.388**
RF, mean AP over five splits													
Abalone19	.029	–	.026	.028	–	–	–	–	–	–	–	–	**.033**
Ecoli4	.828	–	.866	.841	**.925**	.846	.880	.878	.884	.859	.910	.863	.882
page-blocks0	.925	–	.920	.924	.915	.902	.917	.921	.912	.925	.909	–	**.926**
yeast3	.868	–	.851	**.876**	.850	.836	.849	.858	.810	.860	.808	.849	.859
yeast4	.387	–	.364	.349	.352	.332	.421	.403	**.435**	.382	.368	.355	.407
vehicle0	**.984**	–	.982	.983	.931	.951	.974	.982	.980	.980	.975	.979	–
segment0	1.000	–	1.000	1.000	1.000	.992	1.000	1.000	1.000	1.000	1.000	**1.000**	1.000
ecoli1	.833	–	.864	.853	.738	**.875**	.848	.855	.800	.829	.846	.844	.820
abalone9-18	.545	–	**.557**	.522	–	–	–	–	–	–	–	–	.554

Bold: best in row. –: unavailable (no nominal path, library error or no structurally complete candidate; CW-LR is logistic only). SMOTE-ENN/Tomek may remove rows; final counts are in results.

TABLE XIII

EXPLORATORY ADDENDUM, DECLARED AFTER THE MAIN STUDY. AVERAGE RANKS (1 = BEST) OF MEAN F1 AND AP ACROSS DATASETS, WITH EXPLORATORY FRIEDMAN TESTS, AND WILCOXON SIGNED-RANK TESTS OF QUALITY-COVERAGE AGAINST EACH METHOD. NO MULTIPLICITY ADJUSTMENT OR SIGNIFICANCE CLAIM.

(a) Curators, 8 complete datasets

Method	LR F1	LR AP	RF F1	RF AP
Schema rand.	5.12	4.81	4.81	5.00
Adm. rand.	5.12	5.12	4.50	5.12
Source order	4.38	5.38	4.00	4.62
Farthest-first	4.69	3.50	3.62	4.25
Cov.-greedy	3.81	5.00	3.88	5.88
QC (ours)	3.75	4.88	4.06	4.25
CLLM-style†	5.12	2.44	5.88	4.00
Disc.-realism†	4.00	4.88	5.25	2.88
Friedman p	0.654	0.207	0.525	0.387

(b) All methods, 5 complete datasets

Method	LR F1	LR AP	RF F1	RF AP
QC (ours)	6.10	9.40	7.60	8.60
Adm. rand.	10.30	9.80	10.80	9.00
Farthest-first	7.30	7.60	6.60	6.80
CLLM-style†	8.90	6.80	11.20	7.80
Disc.-realism†	5.70	7.60	9.80	4.40
Real only	7.30	7.40	12.50	9.00
ROS+b	9.80	9.20	10.00	8.60
SMOTE(NC)+b	8.30	7.00	10.70	10.20
SMOTE-ENN full	8.50	13.20	7.20	10.40
SMOTE-ENN +b	9.70	7.60	13.00	12.00
SMOTE-Tomek full	7.20	10.20	3.70	7.80
SMOTE-Tomek +b	8.10	5.40	9.80	7.00
B-SMOTE full	10.00	9.00	6.00	7.60
B-SMOTE +b	10.20	5.60	6.50	8.80
ADASYN full	10.80	9.90	4.50	8.20
ADASYN +b	7.80	10.30	6.10	9.80
Friedman p	0.904	0.578	0.031	0.826

(c) Wilcoxon, QC minus method: datasets QC better/worse, p

Method	LR F1	LR AP	RF F1	RF AP
Schema rand.	4/1, 0.12	4/4, 0.64	4/2, 0.31	5/3, 0.95
Adm. rand.	4/1, 0.19	3/4, 0.58	6/2, 0.31	5/3, 1.00
Source order	4/1, 0.44	4/3, 0.81	3/4, 0.58	4/4, 0.84
Farthest-first	4/1, 0.62	3/4, 0.22	3/3, 0.84	5/3, 0.74
Cov.-greedy	1/2, 1.00	3/4, 0.58	2/4, 1.00	5/3, 0.95
CLLM-style†	4/2, 1.00	2/6, 0.08	5/2, 0.38	3/5, 0.84
Disc.-realism†	2/3, 0.81	4/4, 0.46	4/3, 0.94	3/5, 0.38
Real only	5/1, 0.44	3/5, 0.46	5/2, 0.08	5/3, 0.46
CW-LR	6/2, 0.31	3/5, 0.74	–	–
ROS+b	5/1, 0.06	6/2, 0.38	5/1, 0.16	5/3, 0.38
SMOTE(NC)+b	5/0, 0.06	3/5, 0.64	4/2, 0.56	6/2, 0.31
SMOTE-ENN full	4/2, 0.44	5/1, 0.09	3/3, 0.69	4/2, 0.31
SMOTE-ENN +b	3/1, 0.62	3/3, 1.00	5/1, 0.16	5/1, 0.31
SMOTE-Tomek full	4/2, 0.44	4/2, 0.31	2/4, 0.31	3/3, 0.84
SMOTE-Tomek +b	2/2, 1.00	2/4, 0.16	2/3, 0.81	4/2, 0.56
B-SMOTE full	4/2, 0.44	3/3, 0.56	3/3, 1.00	3/3, 0.69
B-SMOTE +b	4/2, 0.56	2/4, 0.44	2/4, 1.00	3/3, 0.69
ADASYN full	5/1, 0.16	4/2, 0.22	2/4, 0.56	3/3, 0.69
ADASYN +b	3/2, 0.81	3/2, 1.00	2/3, 1.00	3/2, 0.62

DECLARATIONS

- **Funding:** No funding was received to assist with the preparation of this manuscript.
- **Conflict of Interest / Competing Interests:** The authors declare that they have no conflict of interest.
- **Ethics Approval and Consent to Participate:** Not applicable.
- **Consent for Publication:** Not applicable.
- **Data Availability:** The datasets analysed in this study are publicly available from the KEEL imbalanced dataset repository [30]; its URL is provided in the reference section.
- **Materials Availability:** Not applicable.
- **Code Availability:** The code used for the experiments is available from the authors on request.
- **Author Contributions:** Dev Sangam Kumar, Aayush Pachpute and Manan Jariwala contributed to the conceptualization, methodology, data analysis, and preparation of the original draft. Priyanka Bhatt contributed to the critical review, editing, supervision, and overall refinement of the manuscript. All authors read and approved the final manuscript.